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QUIZ 3

TRANSPORT PHENOMENA II

CLOSED BOOK, TIME ALLOWED: 40 MIN

Consider a porous cylindrical catalyst with radius $r=R$ and length $z=L$. Species A diffuses into the catalyst. The zeroth-order reaction rate coefficient is k_0'' (moles of A generated per unit time per unit area is $-k_0''$). At the catalyst surface, concentration of A is specified as $c_A(R) = c_{A0}$. Assume:

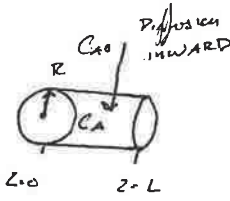
- Stagnant system (neglect convection), $C_A \ll 1$
- Diffusion only happens in the radial direction, $D = D(r)$
- Steady state & isothermal conditions.

$$S = \frac{V}{A}$$

$$S \cdot A = V$$

$$\frac{S \cdot V}{A} = \frac{1}{A}$$

$$\frac{1}{A} = \frac{1}{A}$$



- Given that the mole fraction of A in the catalyst is very small ($x_A \ll 1$), simplify the combined mole flux

$$\langle N_A \rangle = c_A \langle v_A \rangle = c_A \langle v^* \rangle + \langle J_A^* \rangle = c_A \langle v^* \rangle - c D_A \langle \nabla x_A \rangle$$

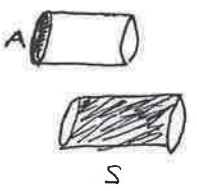
multiply total conc w/ mole fraction to get C_A

$$\langle N_A \rangle = -D_A \langle \nabla C_A \rangle$$

to

Recall: $\langle J_A^* \rangle = -c D_A \langle \nabla x_A \rangle$ where c is the total concentration.

$$\langle N_A \rangle = -D_A \langle \nabla C_A \rangle$$



- Set up a shell balance to derive an ODE for the steady-state concentration profile of A as a function of r

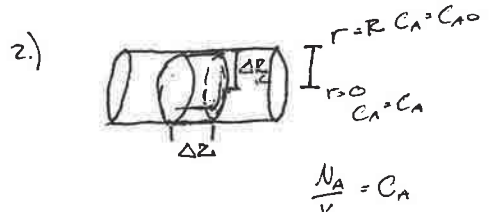
$$\Delta V \left(\int_V dV \frac{\partial C_A}{\partial t} = - \int_S dS \langle N_A \rangle_n + \int_V dV R_A \right)$$

BC2 @ $r=0$ $C_A = C_A$ ~~partial~~

BC1 @ $r=R$ $C_A = C_{A0}$

- List 2 boundary conditions.

- Solve the ODE and determine the integration constants.



Your answers may be in terms of the following constants:

R and L

k_0'' : surface reaction rate constant in moles per unit time per unit area

α : available surface area per unit volume of the catalyst.

D_A : effective diffusivity of species A in the porous medium, where Fick's law in terms of effective diffusivity is

c_{A0} : molar concentration of A at the surface of the catalyst.

$$\left(\frac{\pi r^2 \Delta z}{\pi r^2} \right) \langle N_A \rangle_n = \left(\frac{\pi R^2 \Delta z}{\pi R^2} \right) (R_A)$$

$$\int \frac{\pi r^2 \Delta z}{\pi r^2} - D_A (C_A) = \int \pi R^2 \Delta z (R_A)$$

$$= \pi R^2 \int \Delta z (R_A)$$

$$\int dS - D_A (\Delta C_A) = 0$$

$$\frac{\pi R^2 \Delta z}{\pi R^2} \left(-D_A C_A \right) = 0$$

$$BC1 \quad r=R, \quad C_A = C_{A0}$$

$$C_2 + \frac{6\pi R^2 \Delta z}{6\pi R^2 \Delta z} (-D_A C_{A0}) = 0 \quad C_2 = \frac{6\pi R^2 \Delta z}{6\pi R^2 \Delta z} (-D_A C_{A0})$$

$$0 = - \int dS \langle N_A \rangle_n + \int_V dV R_A$$

$$\int dS \langle N_A \rangle_n = \int_V dV R_A$$

$$\frac{\pi R^2 \Delta z}{\pi R^2} \langle N_A \rangle_n = \frac{\pi R^2 \Delta z}{\pi R^2} (R_A)$$

$$= \pi R^2 \Delta z (R_A)$$

